4.5.13

EE25BTECH11011 - Navya

Question:
Find the equations of the lines parallel to axes and passing through $(2, 3)$
Solution:
The given point is

$$A = \begin{pmatrix} 2 \\ 3 \end{pmatrix}. \tag{1}$$

Case 1: Line parallel to the x-axis
A line parallel to the x-axis has direction vector

$$\mathbf{d} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \tag{2}$$

A normal vector to this line is therefore

$$\mathbf{n} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{3}$$

Equation of a line in normal form is

$$\mathbf{n}^T (\mathbf{x} - A) = 0. \tag{4}$$

Substituting,

$$\begin{pmatrix} 0 & 1 \end{pmatrix} \left(\mathbf{x} - \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right) = 0 \tag{5}$$

$$\begin{pmatrix} 0 & 1 \end{pmatrix} \mathbf{x} = 3 \tag{6}$$

Case 2: Line parallel to the y-axis
A line parallel to the y-axis has direction vector

$$\mathbf{d} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{7}$$

A normal vector to this line is therefore

$$\mathbf{n} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \tag{8}$$

Equation of the line:

$$\begin{pmatrix} 1 & 0 \end{pmatrix} \left(\mathbf{x} - \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right) = 0 \tag{9}$$

$$\begin{pmatrix} 1 & 0 \end{pmatrix} \mathbf{x} = 2 \tag{10}$$

Case 3: General line through $(2, 3)$
Let the normal vector be

$$\mathbf{n} = \begin{pmatrix} a \\ b \end{pmatrix}, \quad (a, b) \neq (0, 0). \tag{11}$$

Equation:

$$\mathbf{n}^T (\mathbf{x} - A) = 0 \tag{12}$$

$$\begin{pmatrix} a & b \end{pmatrix} \left(\mathbf{x} - \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right) = 0 \tag{13}$$

$$\begin{pmatrix} a & b \end{pmatrix} \mathbf{x} = 2a + 3b \tag{14}$$